

Connect all the GPIO pins to the inputs of a Darlington IC ULN2003 and you will be able to control relays connected to the outputs of the IC. The relays, in turn, will be able to switch ACs, fans, lights, curtains, etc on and off. As long as you can access this page with LAN or any other wireless

network, you will be able to control the GPIO pins and the devices attached to them.

Similarly, click on 'OUT' and it will change to 'IN' button, indicating that the port has changed from an output port to an input port.

Interface temperature sensor

DS18B20 digital temperature sensor can sense temperature from -10 to 85 degree celsius. The sensor looks like a small transistor with its three legs, as shown in Fig. 8.

Fig. 8 also shows how to connect the GPIO of Raspberry Pi to temperature sensor DS18B20. Pin 1 (ground) of DS18B20 is connected to pin 6 (ground) of Raspberry Pi. Pin 2 (data) of DS18B20 is connected to pin 7 (GPIO4) of Raspberry Pi. GPIO4 is the only pin of Raspberry Pi which is used as general-purpose clock; any other GPIO pin will not work here. Pin 3 of DS18B20, which is for 3.3-volt supply, will go to pin 1 of Raspberry Pi, which has on-board 3.3-volt supply.

Ensure that a small resistor (4.7k to 10k) is connected between data pin and power pin, as shown in Fig. 8, to pull up data out signal for better correction. Switch on your Raspberry Pi and touch the sensor with a finger to see that it is not getting heated up excessively due to some wrong connection.

After login, load

the DS18B20 module, and the temperature data can be viewed by using command (also refer Fig. 9):

```
# sudo modprobe w1-gpio
# sudo modprobe w1-therm
```

Write the command given below (also refer Fig. 10); you will find the ID number of the DS18B20 sensor as 28-00004ee2c8a:

```
# ls /sys/bus/w1/devices/
```

Your temperature sensor's ID would be different but similar. Now open the webiopi config file using the command:

```
# sudo nano /etc/webiopi/config
```

Go to the [DEVICE] section and add the following lines (also refer Fig. 11):

```
Attic-room-temp = DS18B20 slave:28-00004ee2c8a
tmp0 = DS18B20
```

The first line is for other than DS18B20 sensors, mostly the Chinese sensors. This way they will act as DS18B20 sensors. The second line (tmp0=DS18B20) is for DS18B20 sensors. It does not require any other argument. Both the commands will show the same temperature.

Next, reboot your Raspberry Pi, go to the WebIOPI main menu page and click on Devices Monitor link. You will see the temperature, as shown in Fig. 12.

Interface GPS receiver

You can even interface a GPS receiver and read the data from it on a web browser. Insert your GPS receiver into the USB port of Raspberry Pi and uncomment the following lines in the [DEVICES] section of the webiopi config file:

```
# USB serial adapters
usb0 = Serial device:tttyUSB0
baudrate:9600 //uncomment this line
```

Open the serial monitor on the WebIOPI menu and the GPS data starts pouring in. Just go through the /etc/webiopi/config file and you will come to know many other features of webiopi.

The author is an avid user of open source software. Professionally, he is a thermal power expert and works as an additional general manager at NTPC Limited

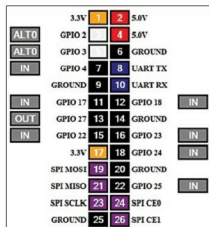


Fig. 7: GPIO header pins

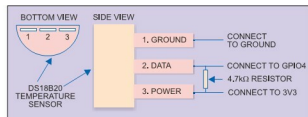


Fig. 8: Pin diagram of DS18B20



Fig. 9: Loading the DS18B20 module



Fig. 10: Finding the ID of temperature sensor

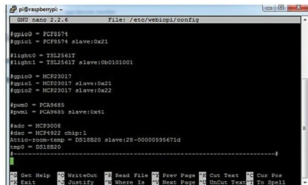


Fig. 11: WebIOPI config file



Fig. 12: Temperature display